



Electromagnetic field penetration in an adult and a child

3. Avoid carrying your cellphone in your pocket, on your belt or in your hand. If you do carry the cellphone in your pocket or hand bag, always position it such that the keypad faces towards you. This way the antenna faces away from you. Body tissue in the abdomen absorbs radiation more quickly than the head. The hip produces 80 per cent of the body's red blood cells and is especially vulnerable to electromagnetic radiation damage. Close proximity may also damage fertility.

4. Consider using your laptop computer to make calls or send an instant message.

5. Avoid using wired headsets. Headsets including the ear buds that come with most cellphones today, act as an antenna, channeling electromagnetic radiation directly to the ear canal. When you use regular ear buds or ear pieces, you get three times more EM radiation than if you held the cellphone against your ear. Also, you get it directly into your ear canal, and ultimately to your brain. So it is recommended not to listen to music on a cellphone using headsets.

6. Use air-tube headsets with ferrite beads. Air-tube headsets are believed to be safer because these don't work as antenna and EM radiation can't travel up the air tube like it does on a regular wired earphone. Further, ferrite beads suppress EM radiation and dissipate it as heat. So, it is recommended to use airtube headsets with a ferrite bead placed as close as possible to the cellphone.

7. Use the speaker phone as much

as possible. EM radiation decreases in direct proportion to the distance of the source from your body. So never hold the phone directly against your head.

8. Avoid making calls when traveling fast. Cellphones automatically increase output power when moving fast as these attempt to connect or handover to the next cellphone base station.

9. When purchasing a cellphone, get one with low SAR.

10. Avoid using a cellphone in metal enclosures like lifts, elevators, vehicles, subways and airplanes. Metal enclosures act like Faraday cage, trapping some of the radiation and reflecting it back upon you and others. So turn off your cellphone inside all metal enclosures.

11. Make calls when and where you have a strong signal. In most systems, the cellphone and the base station check the reception quality and signal strength, and increase or decrease the power level automatically, within a certain span, to accommodate for different situations such as inside or outside buildings and vehicles. When the signal is weak, cellphones automatically increase the power output, exposing you to greater EM radiation.

12. Limit the amount of time you talk on cellphones. Just a two-minute call on a cellphone may disturb the electrical activity of the brain for up to an hour. The more time you spend on mobile phone, the greater your risk of developing brain cancer. So keep your talking brief. You may be saving not only money but a life—quite possibly your own.

13. Use hands-free kit to decrease the radiation to the head.

14. Do not use telephone in a car without an external antenna.

15. The base of a cordless phone emits high levels of EM radiation, even when it is not being used. So use a regular phone as much as possible. Regular phones are the safest bet for conversation and work.

16. Avoid using a cellphone when you are pregnant. The developing organs of the fetus are most sensitive to any possible EM radiation exposure. Moreover, cellphones generate low levels of non-ionising radiation, which are more likely to cause behavioural issues in children after birth. High amounts of these waves can disrupt body tissues—that's one reason why it is suggested to wear protective shielding when getting an X-ray done.

17. Don't allow children to use a cellphone, except for emergencies. Children are at a greater risk because their skulls are not completely thickened. Children's skulls are thinner and their brain contains more fluid than adults'. Exposure of young children to cellphone radiation may be more detrimental to their health, especially during the development and maturation of the central nervous system, immune system and critical organs. Electromagnetic fields easily penetrate through the brain of children, increasing the risk of cancer (see the figure). The electromagnetic field penetration in an adult and a child clarifies it.

18. Use a Bluetooth headset in order to minimise EM radiation. It has been found that Bluetooth headsets emit 1/100th the EM radiation of a normal cellphone. But whatever is radiated by this type of headset directly enters the ear canal and ultimately disturbs the electrical activity of the brain.

19. Use a cellphone radiation protection device which has a patented shield technology that superimposes a low-frequency noise-field onto the radio frequency to make it harmless to the body. ●

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